

23.1.1 Summary of Optimization & Dynamic Optimization so far

23. 1.1.1 Optimisation Problem

Here, we start our summarization of the topic by recalling that optimization problem has three components: *an objective function, decision variables, and constraints*, where

The objective function, often denoted as f or z , reflects a single quantity to be either maximized or minimized.

The decision variables ((also called **choice variables**, often denoted as the **vector $\bar{x}$**) are independent variables, and reflect aspects of the problem that we (or the decision maker) have control over.

Constraints represent any kind of limitation/bounds on the values that the decision variables must satisfy.

The involved concepts are illustrated through the following two examples.

Example 1: A business firm may seek to **maximize profit P** . In the simplest case, profit is the difference between *total revenue R* and *total cost C* ., respectively depending on market demand for the firm's product and production capacity for output, say Q . Thus, both R and C are functions of Q , and hence so is the profit P .

$$P(Q) = R(Q) - C(Q) \quad \dots \dots \dots (I-1)$$

This equation constitutes the relevant objective function, with **P as the object of maximization**

and **Q as the (only) choice variable**.

The optimization problem then is that of choosing the level of Q that maximizes P

It may be noted that the goal is to find the value of Q , for which P is maximized. The goal is *not* to find the maximum value of the choice/independent variable Q .

Example 2: We have 60 feet of fence available, and wish to enclose the **largest possible rectangular area**. What dimensions should be chosen for the fenced-off area? **For this problem,**

The objective is to maximize the rectangular area enclosed by the fence. (Objective function, which is to be maximized, is area bounded by the fence)

The decision variables are not directly given in this problem. Rather, you are told to enclose a rectangular area, which, of course, is determined by **its length (L) and its width (W)**. These are both decision variables which can be controlled directly by us.

There is one obvious constraint—the perimeter of the fence cannot exceed 60 feet—and

two less obvious ones: the length and width must be nonnegative.

All of this information is mathematically represented in the following way:

$$\begin{aligned} \max_{L, W} \quad & L \times W, \\ \text{such that} \quad & 2L + 2W \leq 60; L \geq 0; W \geq 0. \end{aligned} \quad \dots \dots \dots (I-2)$$

23.1.1.2. Dynamic Optimisation problem for continuous choice/independent variable. The form the dynamic optimisation problem, takes, is explained through the following example

Example 3 Consider a profit-making firm which wishes to maximize its profit over a period of time. Let the beginning of time of calculation be t_0 , and end time be t_f . The current profit V depends on both current output and rate of change of output with respect to time. If $x(t)$ denotes output at time t , then the rate of change of output is $x'(t)$, and $V(x(t), x'(t), t)$ denotes the profit *functional*. Thus, *the problem may be mathematically modelled/expressed as*

$$\begin{aligned} \text{Max} \int_{t_0}^{t_f} V(x(t), x'(t), t) dt. \\ \text{Subject to } x(t) \geq 0 \quad x(t_0) = x_0. \end{aligned} \quad \dots \dots \dots (I-3)$$

In general, the *standard problem of optimisation for continuous variable* takes the form of optimizing/ maximizing the following functional

$$V[y] = \int_0^T F[t, y(t), y'(t)] dt \quad \dots \dots \dots (I-4)$$

Where $y=y(t)$, is time-dependent **state-variable**.

In the previous unit, we have mentioned that for handling this type of problems, there are mainly *three dynamic optimization approaches*, viz. (i) calculus of variations, (ii) optimal control theory, which is modern generalization of variational calculus and (iii) dynamic programming.

However, **before we recall**, from the previous unit, our discussion of Calculus of Variations and discuss in detail ‘Optimal Control Theory’, the subject matter of this unit, **we need to discuss, the concept of STATE, which is fundamental/ indispensable concept of any dynamic system.**

23.1.1.3 Concept of STATE in a system & State-space representation

The concept of STATE is quite crucial for understanding the behaviour of a Dynamical System, including of any economic system. However, explaining, what to talk of defining, the concept is hardly attempted, specially, in view of the difficulty involved in doing so. The concept is taken as if it is an *undefined term* like a point, or a straight line in Geometry. **We attempt to explain and define the concept through the following two examples.**

Example 4: Suppose distance between Madurai and Delhi is 2000 Kms. During each cycle of driving, a motorist covers integral multiples of 100 kms, including 0 kms. In order to know how much distance remains to reach Madurai after a number of cycles, the problem is better modelled by using an internal/**system variable** s for recording the *distance covered just before the beginning of the current cycle*, **instead of** keeping record of the sequence of **(cycle-i, distance-i)**, $i = 1, 2, \dots, k$, where distance-i, which—denotes the distance covered during cycle-i—may be 0 kms, 100 kms, 200kms., 300kms,

A **single value**—of the variable s , the **distance-covered-so-far**, with value, say, s_3 denoting **300 kms** covered so far (in general s_i denoting $(100 \times i)$ kms covered so far, $i = 0, 1, 2, \dots$)—**represents various possible combinations of (cycle-i, distance-i):** $(1, 0) + (2, 200) + (3, 100); (1, 100) + (2, 0) + (3, 0) + (4, 100) + (5, 100); (1, 100) + (2, 400); \dots$ Number of such combinations is potentially infinite.

The variable s , the **Distance-covered-so-far** indicates the **progress made toward the goal** of reaching Madurai, **irrespective of** different combinations of **(cycle_i, distance-covered_i)**. From this internal variable and the initial information that Madurai is 2000 Kms from Delhi, we can calculate the required output: the remaining distance. Such an internal variable is called **state-variable with possible values, in this case, as $s_0, s_1, \dots, s_{20}$** . **Using this state-representation of the problem, the following easily follow:** $\text{next-state}(s_4, (\text{cycle}10, 300)) = s_7$; and distance-remining $(s_4, (\text{cycle}10, 300)) = 2000 - (400 + 300) = 1300$ Kms, where s_4 denotes distance-covered so far is 4×100 kms.

The next example, we consider, is about **global system of petroleum production and consumption**. This example will be used in explaining another important concept, viz. **control variable**, which distinguishes the to-be-discussed approach of '**Optimal control**' from the earlier discussed '**Calculus of Variations**' approach.

Example 5: The petroleum production process involves a number of **stages** including exploration, drilling, extraction, processing, storing, transportation, refining and distribution, all before consumption of the product starts.

Each **stage** involves a number of variables, for example, for keeping record of the **extraction process/ stage**, the two variables *amount-of-oil-drilled-out* and *amount-of-oil-extracted* may be required. Thus, there may be a large number of internal variables involved in the complete process from exploration to distribution. However, **if the purpose is to study the consumption pattern of petroleum products over a period of time, only two variables as state-variables**—viz. *sum-amount-of oil-inflow-at-distribution-centre-i* and *sum-amount-of-oil-outflow-at-distribution-centre-i* are required, and—**would be sufficient**.

Explanation & attempted definition of STATE: A *state of a system may be considered* as a sort of indicator/measure of the progress in the direction of the goal/ solution, irrespective of the moves made so far for solving the problem.

In general, **SV, the state-variable**, of a system may consist of more than one internal variable, where an internal variable is a variable which neither appears in the input, nor in the output. Thus, in general, the state-variable is a **state-vector** $(y_1, \dots, y_i, \dots, y_n)$, with y_i 's as internal variables. Particular value of state-vector is called (just) **state** at some particular juncture.

Formally, **SV, the state-vector** is a subset $(y_1, \dots, y_i, \dots, y_n)$ of (possibly very large set) **IV of** internal variables, such that

- i) SV is a **minimal** independent subset of IV, i. e. if an internal variable $z = 3y_1 + 4y_2$, then z is not included in SV
- ii) the **output**: $SV \times IP \rightarrow OP$ is well-defined, i. e. is a function: for each $s_i \in SV$, and $j \in IP$, output $(s_i, j) \in OP$, is unique,
- iii) the **Next-state**: $SV \times IP \rightarrow SV$ is well-defined, i. e. is a function: for each $s_i \in SV$, and $j \in IP$, the next-state $(s_i, j) = s_k$ is unique. We may note that the Next-state function is required to know the outputs corresponding to the later inputs.
- iv) **The state of a dynamical system remains unchanged, unless some input is made from the environment, external to the system.** In detail, though the internal variables may be affecting each other, and the values of some of the internal variables may be changing without external inputs, but the values of those internal variables which constitute the state, remain unchanged between two consecutive inputs.

In essence, STATE variable may be considered as a sort of *implicit, dynamically changing function* from the set of choice-variables to the output, which is

- i) *Dynamically changing*, in the sense, that each given input not only returns the output, but also changes the relation between (state for) the next input and next output, and
- ii) *Implicit*, in the sense, that it is not stated in the form output = f(input), for some f.

State Space Representation: (courtesy Wikipedia): In a dynamic system, including economic systems, a state-space representation is a mathematical model of a physical system specified as a set of input, output and variables, generally related by first-order (not involving second derivatives) differential equations or difference equations. As explained above, such variables, called **state variables**, evolve over time in a way that depends on the values they have at any given instant and on the externally imposed values of input variables. Output variables' values depend on the values of the state variables.

23.1.3 Calculus of variations: Summary

The *standard problem of optimisation for continuous variable*, mentioned earlier, as of optimizing/ maximizing the following functional

$$V[y] = \int_0^T F[t, y(t), y'(t)] dt \quad \dots \dots \dots (I-4)$$

where $y = y(t)$, is time-dependent **state-variable**,

takes the following form, in the case of ‘Calculus of Variations’:

Optimise/ maximize the following functional

$$V[y] = \int_0^T F[t, y(t), y'(t)] dt \quad \dots \dots \dots (I-5)$$

Subject to the conditions that

Initial state $y(0)$ (say =A), the terminal time T , and

the final state $y(T)$ (= say Z) are given.

In the above, $y(t)$ is the state at time t , and $y'(t)$ is the rate of change in state-variable at time t .

The solution of (I-5) is given by the following Euler-Lagrange equations

$$F_y = \frac{d}{dt} F_{y'} = 0 \quad \text{for all } t \in [0, T] \quad \dots \dots \dots (I-6)$$

And, more explicitly by the equation

$$F_{y'y'} y''(t) + F_{yy'} y'(t) + F_{ty'} - F_y = 0 \quad \dots \dots \dots (I-7)$$

23.1.4 Optimal control theory: Summary

We start our discussion of ‘*optimal control theory*’ with mentioning the fact that the essential difference between ‘*Calculus of Variations*’ and ‘*Optimal control Theory*’ is the additional use of ‘**control variable**’ in the latter.

Combined with Example 5 above, in this context, let us recall the recent news (dated: June 04, 2023): OPEC+ member countries consider more production cuts as oil prices slump.

*The above is the case of use of **control variable**: Despite the availability of technology and capacity for production, **the oil will not be produced according to maximum capacity** because of the above decision which stipulates the upper limit on the production over a period of time.*

A control variable is external to the production-consumption system.

With this, we outline the Standard/ simplest Optimal Control Problem and its Theory

$$\text{Maximize or minimize } V[u] = \int_0^T F[t, y(t), u(t)] dt$$

Subject to the conditions that

$$\mathbf{y}'(t) = \mathbf{f}(t, \mathbf{y}(t), \mathbf{u}(t)),$$

initial state $\mathbf{y}(0)$ (say $=A$), the terminal time T , and (I-8)

the final state $\mathbf{y}(T)$ (= say Z) are given.

The equation $\mathbf{y}'(t) = \mathbf{f}(t, \mathbf{y}(t), \mathbf{u}(t))$, derivative of $\mathbf{y}$ with respect to t is called **State Equation** as well as **Equation of Motion**.

Though (I-8) appears to be quite similar to (I-5), yet there are subtle and significant differences in the optimization problems corresponding to respectively 'Optimal Control Theory' and 'Calculus of Variations', as discussed below

- i) $\mathbf{y}'(t)$ = the rate of change of the state in (I-5) was a function of t and $\mathbf{y}(t)$ only. But in 'Optimal Control Theory', $\mathbf{y}'(t) = \mathbf{f}(t, \mathbf{y}(t), \mathbf{u}(t))$, i. e. it is also **controlled by the (external) control variable. This is a significant change.**
- ii) The functional to be optimized in 'Calculus of Variations' (i. e., in (I-5)), is $V(\mathbf{y})$, i. e. it is treated mainly as a function of $\mathbf{y}$, **the internal state**. But the functional to be optimized in 'Optimal Control Theory' (i. e., in (I-8)), is $V(\mathbf{u})$, i. e. it is treated mainly **as a function of $\mathbf{u}$, the external control variable**. In other words, the **emphasis in the problem definition** has shifted from the **internal state $\mathbf{y}$** to **$\mathbf{u}$, the external control variable**.

23.1.4.1 Slight Modification in notation

In order to keep uniformity of notation used in the rest of this unit, the following changes to the above notations have been made:

- i) The objective function, for which we used $V(\mathbf{u})$, henceforth, we **will be** using $J(\mathbf{u})$
- ii) The state variable for which we used $\mathbf{y}(t)$, henceforth, we will be using $\mathbf{x}(t)$
- iii) The derivative with respect to t will be denoted as $\dot{\mathbf{x}}$, instead of $\mathbf{x}'$ etc.
- iv) The initial time is taken as $t = t_0$ instead of $t = 0$. And the final time will be taken as t_f instead of T .

In view of the above notational changes, Standard/ simplest Optimal control problem represented by (I-8), now becomes

$$J(\mathbf{u}) \rightarrow \max_{\mathbf{u}(\cdot)},$$

$$\dot{\mathbf{x}}(t) = \mathbf{f}(t, \mathbf{x}(t), \mathbf{u}(t)),$$

$\mathbf{x}(t_0)$ denotes initial state, and $\mathbf{x}(t_f)$ denotes final state.

23.2 FIRST ORDER NECESSARY CONDITION: EULER LAGRANGE EQUATION

Consider a dynamic system described by the following system of differential equations.

$$\dot{x}(t) = \frac{dx(t)}{dt} = f(x(t)u(t), t) \quad \dots (23.1)$$

with initial condition $x(t_0) = x(0) = x^0$

where $x(t) = [x_1(t), x_2(t), \dots, x_n(t)]^T$

$$u(t) = [u_1(t), u_2(t), \dots, u_m(t)]^T$$

$x(t)$ and $u(t)$ are respectively referred to as state and control variables.

$f = [f_1, f_2, \dots, f_n]^T$ is assumed to be continuously differentiable with respect to all its arguments and x^0 is given vector in P^n

We have assumed that the given process starts at $t = 0$ and ends at the fixed terminal time $t_f > 0$. Any piecewise continuous function u from $[t_0, t_f]$ into P^m may be taken as an admissible control.

We now consider an optimal control problem where a control $u \in U$ is to be chosen such that the cost functional/objective functional/performance index.

$$J(u) = S(x(t), t) \big|_{t=t_f} + \int_{t_0}^{t_f} V(x(t), u(t), t) dt \quad \dots (23.2)$$

is minimized.

Note here that $J(\cdot)$ is a scalar quantity, $S(x(t), t) \big|_{t=t_f}$ is terminal cost and a continuous function.

We assume $S(\cdot)$ and $V(\cdot)$ are continuously differentiable with respect to their respective arguments.

Note that the cost functional can be regarded as depending on u only as x is implicitly determined by u . from (23.1).

Our objective is $\xrightarrow{u(t)}$ System $\xrightarrow{x(t)}$ to find $u(t)$ such that objective functional is optimised. In other words, we have a continuous dynamic optimisation problem at hand with equality constraint.

The popular name of such problem is called “Bolza problem”.

The terminal cost is in quadratic form and must be positive quantity. The constraint is also called dynamic equation of system.

$$\text{Now } \int_{t_0}^{t_f} \left[\frac{ds(x(t), t)}{dt} \right] dt = [S(x(t), t)]_{t=t_0}^{t=t_f}$$

$$= S(x(t_f), t_f) - S(x(t_0), t_0)$$

$$\Rightarrow S(x(t_f), t_f) = \int_{t_0}^{t_f} \left[\frac{ds(x(t), t)}{dt} \right] dt + S(x(t_0), t_0) \quad \dots (23.3)$$

Using equation (23.3) in (23.2) we get

$$J(.) = \int_{t_0}^{t_f} v(x(t), u(t), t) dt + \int_{t_0}^{t_f} \frac{ds(x(t), t)}{dt} dt + S(x(t_0), t_0) \quad \dots (23.4)$$

Because $S(x(t_0), t_0)$ is fixed Quantity, minimization of 1st two integral in above equation give minimal value of functional $J(.)$.

Therefore optimisation of equation (23.4) is equivalent to that of

$$J(.) = \int_{t_0}^{t_f} v(x(t), u(t), t) dt + \int_{t_0}^{t_f} \frac{ds(x(t), t)}{dt} dt \quad \dots (23.5)$$

We know that if there is function

$$f(x(t), y(t), z(t))$$

$$\text{then } \frac{\partial f}{\partial t} = \frac{\partial f}{\partial x} \cdot \frac{\partial x}{\partial t} + \frac{\partial f}{\partial y} \cdot \frac{\partial y}{\partial t} + \frac{\partial f}{\partial z} \cdot \frac{\partial z}{\partial t}$$

$$= \frac{\partial f}{\partial x} x'(t) + \frac{\partial f}{\partial y} y'(t) + \frac{\partial f}{\partial z} z'(t)$$

Hence our problem is to optimise $J(.)$ which becomes

$$J(.) = \int_{t_0}^{t_f} v(x(t), u(t), t) dt + \int_{t_0}^{t_f} \left[\frac{\partial s}{\partial x} \cdot x'(t) + \frac{\partial s}{\partial t} \cdot 1 \right] dt \quad \dots (23.6)$$

$$\text{Subject to } x'(t) = f(x(t), u(t), t) \quad \dots (23.7)$$

We need to find $u^*(t)$ converting equation (23.6) and (23.7) into unconstrained optimisation problem using lagrange multiplier our modified objective functional becomes

$$J(.) = \int_{t_0}^{t_f} \left[v(x(t), u(t), t) dt + \left[\frac{\partial s(x(t), t)}{\partial x(t)} \cdot x'(t) + \frac{\partial s(x(t), t)}{\partial t} \right] dt \right.$$

$$\left. + \lambda(t) [f(x(t), u(t), t) - x'(t)] dt \right] \quad \dots (23.8)$$

What we have done is that instead of $x(t)$ as a vector and $u(t)$ as vector we have taken them as scalar for sake of simplicity. Later it will be generalized to a multivariate case.

Moreover, we have added $\lambda(t)[f(x(t), u(t), t) - x'(t)]dt$ to objective functional because $x'(t) = f(x(t), u(t), t)$, so no change in value of functional.

$\lambda(t)$ is Lagrange multiplier this can be interpreted as marginal valuation of the associated state variable at time t .

Equation 23.8 can be written as

$$\int_{t_0}^{t_f} \Lambda(x(t), u(t), \lambda(t), t) dt$$

Where Lagrangian function $\Lambda(x(t), u(t), \lambda(t), t)$ is equal to

$$v(x(t), u(t), t) + \lambda(t)f(x(t), u(t), t) + \frac{\partial s(x(t), t)}{\partial x(t)} \cdot x'(t) + \frac{\partial s(x(t), t)}{\partial t} - \lambda(t)x'(t)$$

We further write Hamiltonian function

$$H(x(t), u(t), \lambda(t), t) = V(x(t), u(t), t) + \lambda(t)f(x(t), u(t), t) \quad \dots (23.9)$$

Hamiltonian function is named after 19th century Irish mathematician W.R. Hamilton

Therefore, our Lagrangian function becomes

$$\Lambda(x(t), u(t), \lambda(t), t) = H(x(t), u(t), \lambda(t), t) + \left[\frac{\partial s(x(t), t)}{\partial x(t)} \right] \cdot x'(t) + \frac{\partial s(x(t), t)}{\partial t} - \lambda(t)x'(t) \quad \dots (23.10)$$

The question arises, what is the value of perturbed condition. Perturbed condition means the sequence $J(\cdot)$ is defined when x^* is perturbed by $x^*(t) + \delta(x(t))$ and $u(t)$ is perturbed by $u^*(t) + \delta u(t)$. Our job is to look for optimal trajectory of $u^*(t)$ which, in turn, will optimise our functional.

Let that perturbed objective function be $J_a(\cdot)$

Hence, we can write

$$J_a(\cdot) = \int_{t_0}^{t_f + \delta t_f} \Lambda_p(\cdot) dt \quad \dots (23.11)$$

Where $\Lambda_p(\cdot)$ is Lagrangian function of perturbed state. This is achieved by replacing $u(t) = u^*(t) + \delta u_t$ and $x(t) = x^*(t) + \delta x(t)$ in equation (23.10)

$$\Rightarrow J_a(\cdot) = \int_{t_0}^{t_f} \Lambda_p(\cdot) dt + \int_{t_f}^{t_f + \delta t_f} \Lambda_p(\cdot) dt$$

Consider the figure in this page

$\Lambda_p(\cdot)$ is a scalar

If we consider

$$\int_{t_f}^{t_f + \delta t_f} \Lambda_p(\cdot) dt = \text{area under the curve AE, which is nearly equal to area ABDC.}$$

Hence $\int_{t_f}^{t_f + \delta t_f} \Lambda_p(\cdot) dt$ approximately equals to $\Lambda(\cdot) |_{t=t_f} \cdot \delta t_f$, as δt_f is very small.

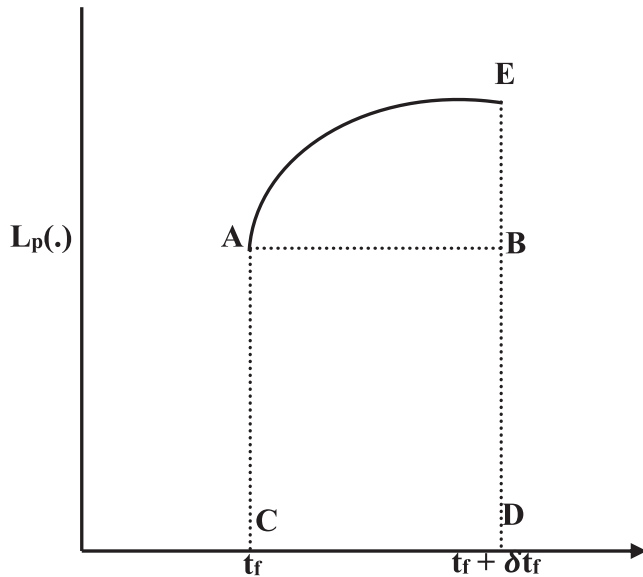


Fig. 23.1: Trajectory of Lagrange Functional

Therefore, variation to functional $\Delta J_a = J_a(.) - J(.)$

$$\begin{aligned}
 \Rightarrow \Delta J_a &= \int_{t_0}^{t_f + \delta t_f} \Lambda_p(.) dt - \int_{t_0}^{t_f} \Lambda(.) dt \\
 &\cong \int_{t_0}^{t_f} \Lambda_p(.) dt + \Lambda(.)|_{t=t_f} \delta t_f - \int_{t_0}^{t_f} \Lambda(.) dt \\
 &\cong \int_{t_0}^{t_f} [\Lambda_p(.) - \Lambda(.)] dt + \Lambda(.)|_{t=t_f} \delta t_f \\
 &\cong \int_{t_0}^{t_f} [\Lambda_p(x^*(t) + \delta x(t), u^*(t) + \delta u(t), \lambda(t), t) \\
 &\quad - \Lambda(x^*(t), u^*(t), \lambda(t), t)] dt \\
 &\quad + \Lambda(x^*(t), u^*(t), \lambda(t), t)|_{t=t_f} \cdot \delta t_f \quad \dots (23.12)
 \end{aligned}$$

Using Taylor series expansion and integration by parts we can get 1st variation of functional (pleas ref to unit -22 for proof of theorem in calculus of variations)

For necessary condition after simplification

$\partial J = 1^{\text{st}}$ variation of functional

$$= \int_{t_0}^{t_f} \left[\frac{\partial \Lambda(.)}{\partial x(t)} - \frac{d}{dt} \left(\frac{\partial \Lambda(.)}{\partial x'(t)} \right) \right]_* \delta x(t) \cdot dt$$

* means at optimal path

$$+ \int_{t_0}^{t_f} \left(\frac{\partial \Lambda(.)}{\partial u(t)} \right)_* \delta u(t) dt + \left(\frac{\partial \Lambda(.)}{\partial x'(t)} \right)_* \delta x(t)|_{t=t_f} + \Lambda(.)|_{t=t_f} \cdot \delta t_f \quad \dots (23.13)$$